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DISCRETE MATHEMATICS

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PART1: COMBINATORIAL THEORY

PART 2: GRAPH THEORY

Contents of part 1

- Chapter 0: Sets, Functions
- Chapter 1: Counting problem
- Chapter 2: Existence problem
- **Chapter 3: Enumeration problem**
- Chapter 4: Combinatorial optimization problem

1. **Introduction to problem**
2. Generating algorithm
3. Backtracking algorithm

1. Introduction to problem

- A problem asking to give a list of all combinations that satisfy a given number of properties is called the combinatorial enumeration problem.
- As the number of combinations to enumerate is usually very large even when the configuration size is not large:
  - The number of permutations of n elements is $n!$
  - The number of subset consisting m elements taken from n elements is $n!/(m!(n-m)!)$

Therefore, it is necessary to have the concept of solving combinatorial-enumeration problems

1. Introduction to problem

- The combinatorial enumeration problem is solvable if we can define ***an algorithm*** so that all configurations could be built in turn.
- An enumeration algorithm must satisfy 2 basic requirements:
  - Do not repeat a configuration,
  - Do not miss a configuration.

1. Introduction to problem
2. **Generating algorithm**
3. Backtracking algorithm

2. Generating algorithm

2.1. Algorithm diagram

2.2. Generate basic combinatorial configurations

- Generate binary strings of length n
- Generate m-element subsets of the set of n elements
- Generate permutations of n elements

2.1. Algorithm diagram

The generating algorithm could be applied to solve the combinatorial enumeration problem if the following two conditions are fulfilled:

1) An order can be specified on the set of combinations to enumerate. From there, the first and last configuration could be defined in the order specified.
2) Build the algorithm to give the next configuration of the current configuration (not the final one yet).

The algorithm referred to in condition 2) is called Successive Generation Algorithm

2.1. Algorithm diagram

```
void Generate ( )
{
  <Build the first configuration>;
  stop=false;
  while (!stop)
  {
       <Print out the current configuration>;
       if (the current configuration is not the final yet)
         Successive_Generation ( );
      else stop = true;
  }
}
```

<Successive_Generation> is the procedure create the next configuration of the current one in the order specified.

Note: Since the set of combinatorial configurations to enumerate is finite, we can always define an order on them. However, the order should be determined so that the successive generation algorithm could be built.

2. Generating algorithm

2.1. Algorithm diagram

2.2. Generate basic combinatorial configurations

- **Generate binary strings of length *n***
- Generate *m*-element subsets of the set of *n* elements
- Generate permutations of *n* elements

2. Generating algorithm

Problem: Enumerate all binary strings of length n:

$$b_1\, b_2 \ldots b_n, \text{ where } b_i \in \{0, 1\}.$$

Solution:

- Dictionary order:

Consider each binary string $b = b_1\, b_2 \ldots b_n$ as the binary representation of an integer number $p(b)$

We say binary string $b = b_1\, b_2 \ldots b_n$ is ***previous*** binary string $b' = b'_1\, b'_2 \ldots b'_n$ in dictionary order and denote as $b \prec b'$ if $p(b) < p(b')$.

2. Generating algorithm

Example: When n=3, all binary strings of length 3 are enumerated in the dictionary order in the table as following:

b	$p(b)$
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

2. Generating algorithm

Successive_Generation algorithm:

- The first string is 0 0 ... 0,
- The last string is 1 1 ... 1.
- Assume $b_1\ b_2 \ldots b_n$ *be the current string*:
  - If this string consists of all 1 → finish,
  - Otherwise, the next string is obtained by adding 1 (module 2, memorize) to the current string.
- Thus, we have following rule to generate the next string:
  - Find the first *i* (in the order *i* = *n*, *n*-1, ..., 1) such that $b_i = 0$.
  - Reassign $b_i = 1$ and $b_j = 0$ for all $j > i$. The newly obtained sequence will be the string to find.

2. Generating algorithm

Example: consider binary string of length 10: b = 1101011111. Find the next string:.

- We have i = 5. Thus, set:
  - b_5 = 1,
  - and b_i = 0, i = 6, 7, 8, 9, 10

→ We obtain the next string is 1101100000.

$$\begin{array}{r} \mathbf{1101011111} \\ +\quad \mathbf{1} \\ \hline \mathbf{1101100000} \end{array}$$

2. Generating algorithm

Successive_Generation algorithm:

```
void  Next_Bit_String( )
/*Generate the next binary string in the dictionary order
of the current string  b1 b2 ... bn ≠ 1 1 ... 1   */
{
    i = n
    while  (b[i] == 1):
        b[i] = 0
        i = i-1
    b[i] = 1
}
```

2. Generating algorithm

2.1. Algorithm diagram

2.2. Generate basic combinatorial configurations

- Generate binary strings of length *n*
- **Generate *m*-element subsets of the set of *n* elements**
- Generate permutations of *n* elements

2. Generating algorithm

Problem: Let $X = \{1, 2, \ldots, n\}$. Enumerate all m-element subsets of X.

Solution:

- Lexicographic order:

Each m-element subset of X could be represented by tuples of m elements

$$a = (a_1, a_2, \ldots, a_m)$$

satisfying

$$1 \le a_1 < a_2 < \ldots < a_m \le n.$$

2. Generating algorithm

We say subset $a = (a_1, a_2,..., a_m)$ is previous subset $a' = (a'_1, a'_2, ... , a'_m)$ in dictionary order and denote as $a \prec a'$, if one could find the index k ($1 \leq k \leq m$) such that:

$$a_1 = a'_1 , a_2 = a'_2, . . . , a_{k-1} = a'_{k-1},$$

$$a_k < a'_k .$$

2. Generating algorithm

Example: Enumerate all 3-element subset of $X = \{1, 2, 3, 4, 5\}$ in dictionary order

1	2	3
1	2	4
1	2	5
1	3	4
1	3	5
1	4	5
2	3	4
2	3	5
2	4	5
3	4	5

2. Generating algorithm

Successive_Generation algorithm:

- The first subset is (1, 2, ... , m)
- The last subset is (n-m+1, n-m+2, ..., n).
- Assume $a=(a_1, a_2, \dots, a_m)$ is the current subset but not the final yet, then its next subset in the dictionary order could be built by using the following rules:
    - Scan from the right to the left of sequence $a_1, a_2, \dots, a_m$: find the first element $a_i \neq n-m+i$
    - Replace a_i by $a_i + 1$
    - Replace a_j by $a_i + j - i$, where $j = i+1, i+2, \dots, m$

2. Generating algorithm

Example: $n = 6, m = 4$

Assume the current subset (1, 2, 5, 6), we need to build its next subset in the dictionary order:

- Scan from the right to the left of sequence $a_1, a_2, \ldots, a_m$: find the first element $a_i \neq n\text{-}m\text{+}i$
- Replace a_i by $a_i + 1$
- Replace a_j by $a_i + j - i$, where $j = i+1, i+2, \ldots, m$

- We have i=2:

Sequence: (1, **2,** 5, 6)

Value n-m+i: (3, **4,** 5, 6)

replace $a_2 = a_2+1 = 3$

$a_3 = a_i + j - i = a_2 + 3 - 2 = 4$

$a_4 = a_i + j - i = a_2 + 4 - 2 = 5$

We then obtain its next subset (1, 3, 4, 5).

Successive_Generation algorithm:

```
void Next_Combination( )
/*Generate the next subset in dictionary order of the subset (a1, a2,...,
  am)  ≠  (n-m+1,...,n)*/
{
    i = m
    while (a[i] == n-m+i):
        i = i-1
    a[i]= a[i] + 1
    for j in range (i+1, m+1):
        a[j] = a[i] + j - i
}
```

2. Generating algorithm

2.1. Algorithm diagram

2.2. Generate basic combinatorial configurations

- Generate binary strings of length *n*
- Generate *m*-element subsets of the set of *n* elements
- **Generate permutations of *n* elements**

2. Generating algorithm

Problem: Give $X = \{1, 2, \ldots, n\}$, enumerate all permutations of n elements of X.

Solution:

- Dictionary order:
  - Each permutation of n elements of X could be represented by an ordered set of n elements:

$$a = (a_1, a_2, \ldots, a_n)$$

satisfy

$$a_i \in X,\ i = 1, 2, \ldots, n,\ a_p \neq a_q,\ p \neq q.$$

2. Generating algorithm

We say permutation $a = (a_1, a_2,..., a_n)$ is previous permutation $a' = (a'_1, a'_2, ... , a'_n)$ in dictionary order and denote as $a \prec a'$, if we could find the index k ($1 \leq k \leq n$) such that:

$$a_1 = a'_1 , a_2 = a'_2, ... , a_{k-1} = a'_{k-1},$$

$$a_k < a'_k .$$

2. Generating algorithm

Example: All permutation of 3 elements of X ={1, 2, 3} could by enumerated in dictionary order as following:

1	2	3
1	3	2
2	1	3
2	3	1
3	1	2
3	2	1

Successive_Generation algorithm:

- The first permutation: $(1, 2, \dots, n)$
- The last permutation: $(n, n\text{-}1, \dots, 1)$.
- Assume $a = (a_1, a_2, \dots, a_n)$ is not the last permutation, then its next permutation could be built using the following rules:
  - Find from the right to left: first index j satisfying $a_j < a_{j+1}$ (in other word: j is the max index satisfying $a_j < a_{j+1}$);
  - Find a_k be the smallest number among those on the right of a_j in the sequence and *greater than* a_j ;
  - Swap a_j and a_k ;
  - Invert the sequence from a_{j+1} to a_n .

2. Generating algorithm

Example: Assume the current permutation (3, 6, 2, 5, 4, 1), need to create its next permutation in the dictionary order.

- Find from the right to left: first index j satisfying $a_j < a_{j+1}$ (in other word: j is the max index satisfying $a_j < a_{j+1}$);
- Find a_k be the smallest number among those on the right of a_j in the sequence and *greater than* a_j ;
- Swap a_j and a_k ;
- Invert the sequence from a_{j+1} to a_n .

- We have index $j = 3$ ($a_3 = 2 < a_4 = 5$).
- The smallest number among those on the right of a_3 and greater than a_3 is $a_5 = 4$. *Swap* a_3 and a_5 in the current permutation (3, 6, **2**, 5, **4**, 1)

  we obtain (3, 6, **4**, 5, **2**, 1),

- Finally, invert the sequence $a_4\ a_5\ a_6$, we obtain the next permutation is (3, 6, 4, **1, 2, 5**).

Successive_Generation algorithm:

```
Next_Permutation ( ){
  /*Generate next permutation  (a1, a2, ..., an)  ≠ (n, n-1, ..., 1)    */
  // Find j being max index satisfying aj < aj+1 :
  j=n-1;  while (aj > aj+1 ) j=j-1;
  // Find ak being smallest number among those on the right aj and greater //than aj :
  k=n;
  while (aj > ak ) k=k-1;
  Swap(aj, ak); //swap aj and ak
  // Invert the sequence aj+1 to an :
  r=n; s=j+1;
  while ( r>s ) {
        Swap(ar, as);  // swap ar and as
        r=r-1; s= s+1;
  }
}
```

1. Introduction to problem
2. Generating algorithm
3. **Backtracking algorithm**

3.1. Algorithm diagram

3.2. Generate basic combinatorial configurations

- Generate binary strings of length *n*
- Generate *m*-element subsets of the set of *n* elements
- Generate permutations of *n* elements

3.1. Algorithm diagram

- ***Enumeration problem* (Q):** *Given A_1, A_2,..., A_n be finite sets. Denote*

  $A = A_1 \times A_2 \times ... \times A_n = \{ (a_1, a_2, ..., a_n): a_i \in A_i , i=1, 2, ..., n\}.$

  Assume P is a property on the set A. The problem is to enumerate all elements of the set A that satisfies the property P:

  $$D = \{ a = (a_1, a_2, ..., a_n) \in A: a \text{ satisfy property } P \}.$$

- Elements of the set D are called **feasible solution** (*lời giải chấp nhận được*).

3.1. Algorithm diagram

The solution to the problem is an ordered tuple of n elements $(a_1, a_2, ..., a_n)$, where $a_i \in A_i$, $i = 1, 2, ..., n$.

Definition. The k-level partial solution ($0 \le k \le n$) is an ordered tuple of k elements

$$(a_1, a_2, ..., a_k),$$

where $a_i \in A_i$, $i = 1, 2, ..., k$.

- When $k = 0$, 0-level partial solution is denoted as ( ), and called as the empty solution.
- When $k = n$, we have a complete solution to a problem.

3.1. Algorithm diagram

Backtracking algorithm is built based on the construction each component of solution one by one.

- Algorithm starts with empty solution ( ).
- Based on the property P, we determine which elements of set A_1 could be selected as the first component of solution. Such elements are called as **candidates** for the first component of solution. Denote candidates for the first component of solution as S_1. Take an element $a_1 \in S_1$, insert it into empty solution, we obtain 1-level partial solution: (a_1).

3.1. Algorithm diagram

- General step: Assume we have k-1 level partial solution:

$$(a_1, a_2, \ldots, a_{k-1}),$$

Now we need to build k-level partial solution:

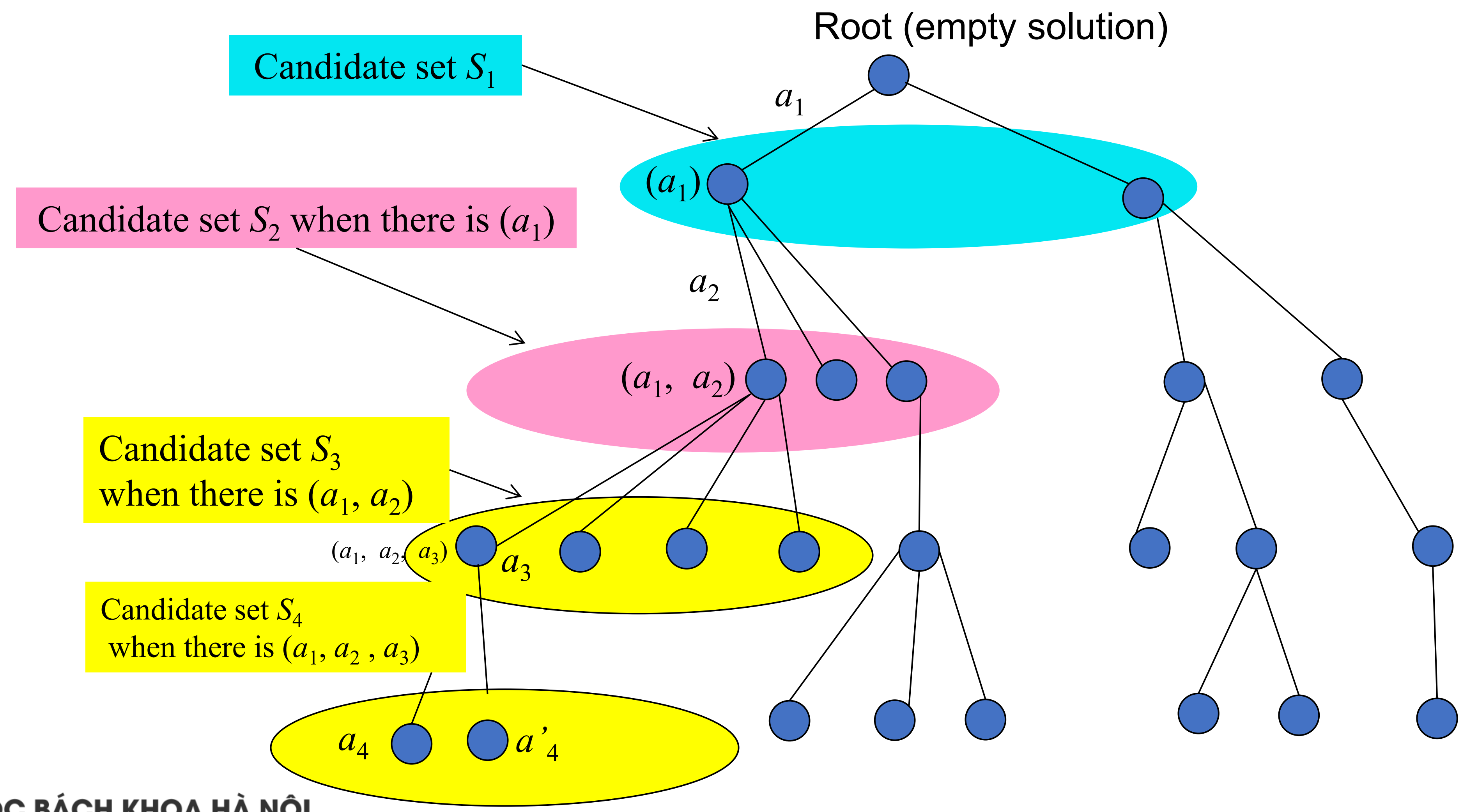
$$(a_1, a_2, \ldots, a_{k-1}, \mathbf{a_k})$$

  - Based on the property P, we determine which elements of set A_k could be selected as the k^{th} component of solution.
  - Such elements are called as candidates for the k^{th} position of solution when k-1 first components have been chosen as $(a_1, a_2, \ldots, a_{k-1})$. Denote these candidates by S_k.
  - Consider 2 cases:
    - $S_k \neq \varnothing$
    - $S_k = \varnothing$

3.1. Algorithm diagram

- **$S_k \neq \varnothing$:** Take $a_k \in S_k$ to insert it into current (k-1)-level partial solution $(a_1, a_2, ..., a_{k-1})$, we obtain k-level partial solution $(a_1, a_2, ..., a_{k-1}, a_k)$. Then
  - If $k = n$, then we obtain a complete solution to the problem,
  - If $k < n$, we continue to build the $(k+1)^{th}$ component of solution.
- **$S_k=\varnothing$:** It means the partial solution $(a_1, a_2, ..., a_{k-1})$ can not continue to develop into the complete solution. In this case, we **backtrack** to find new candidate for $(k\text{-}1)^{th}$ position of solution (note: this new candidate must be an element of S_{k-1})
  - If one could find such candidate, we insert it into $(k\text{-}1)^{th}$ position, then continue to build the k^{th} component.
  - If such candidate could not be found, we backtrack one more step to find new candidate for $(k\text{-}2)^{th}$ position,... If backtrack till the empty solution, we still can not find new candidate for 1^{st} position, then the algorithm is finished.

Decision tree for backtracking



3.1. Algorithm diagram

Backtracking algorithm (recursive)

```
void Try(int k)
{
    <Build Sk as the set to consist of candidates for the
  kth component of solution>;
   for y ∈ Sk  //Each candidate y of Sk
   {
  ak = y;
       if  (k == n) then <Record (a1, a2, ..., ak) as a
  complete solution >;
  else Try(k+1);
       Return the variables to their old states;
  }
}
```

The call to execute backtracking algorithm: Try(1);

- If only one solution need to be found, then it is necessary to find a way to terminate the nested recursive calls generated by the call to Try(1) once the first solution has just been recorded.
- If at the end of the algorithm, we obtain no solution, it means that the problem does not have any solution.

Backtracking algorithm (not recursive)

```
void Backtracking ( )
{
  k=1;
  <Build Sk>;
  while (k > 0)  {
     while (Sk ≠ ∅ ) {
        ak ← Sk; // Take ak from Sk
        if <(k == n) > then <Record (a1,a2,...,ak) as a
                                  complete solution >;
        else {
              k = k+1;
              <Build Sk>;
        }
     }
     k = k - 1;   // Backtracking
  }
}
```

The call to execute backtracking algorithm: Backtracking ( );

3.1. Algorithm diagram

2 keys issues: In order to implement a backtracking algorithm to solve a specific combinatorial problems, we need to solve the following two basic problems:

- Find algorithms to build candidate sets S_k
- Find a way to describe these sets so that you can implement the operation to enumerate all their elements (implement the loop for $y \in S_k$).
  - The efficiency of the enumeration algorithm depends on whether we can accurately identify these candidate sets.

3.1. Algorithm diagram

- If the length of complete solution is not known in advanced, and solutions are not needed to have the same length:

```
void Try(int k)
{
   <Build Sk as the set to consist of candidates for the kth component of solution>;
  for y ∈ Sk  //Each candidate y of Sk
  {
     ak = y;
     if  (k == n) then <Record (a1, a2, ..., ak) as a complete solution >;
     else Try(k+1);
     Return the variables to their old states;
  }
}
```

- Then we need to modify statement

if (*k* == *n*) then <Record (a_1, a_2, ..., a_k) as a complete solution >;

else Try(k+1);

to

if <(a_1, a_2, ..., a_k) is a complete solution> then <Record (a_1, a_2, ..., a_k) as a complete solution >;

else Try(k+1);

→ Need to build a function to check whether (a_1, a_2, ..., a_k) is the complete solution.

3. Backtracking algorithm

3.1. Algorithm diagram

3.2. Generate basic combinatorial configurations

- **Generate binary strings of length *n***
- Generate *m*-element subsets of the set of *n* elements
- Generate permutations of *n* elements

Generate binary strings of length *n*

- Problem to enumerate all binary string of length *n* leads to the enumeration of all elements of the set:

  $$A^n = \{(a_1, ..., a_n): a_i \in \{0, 1\}, i=1, 2, ..., n\}.$$

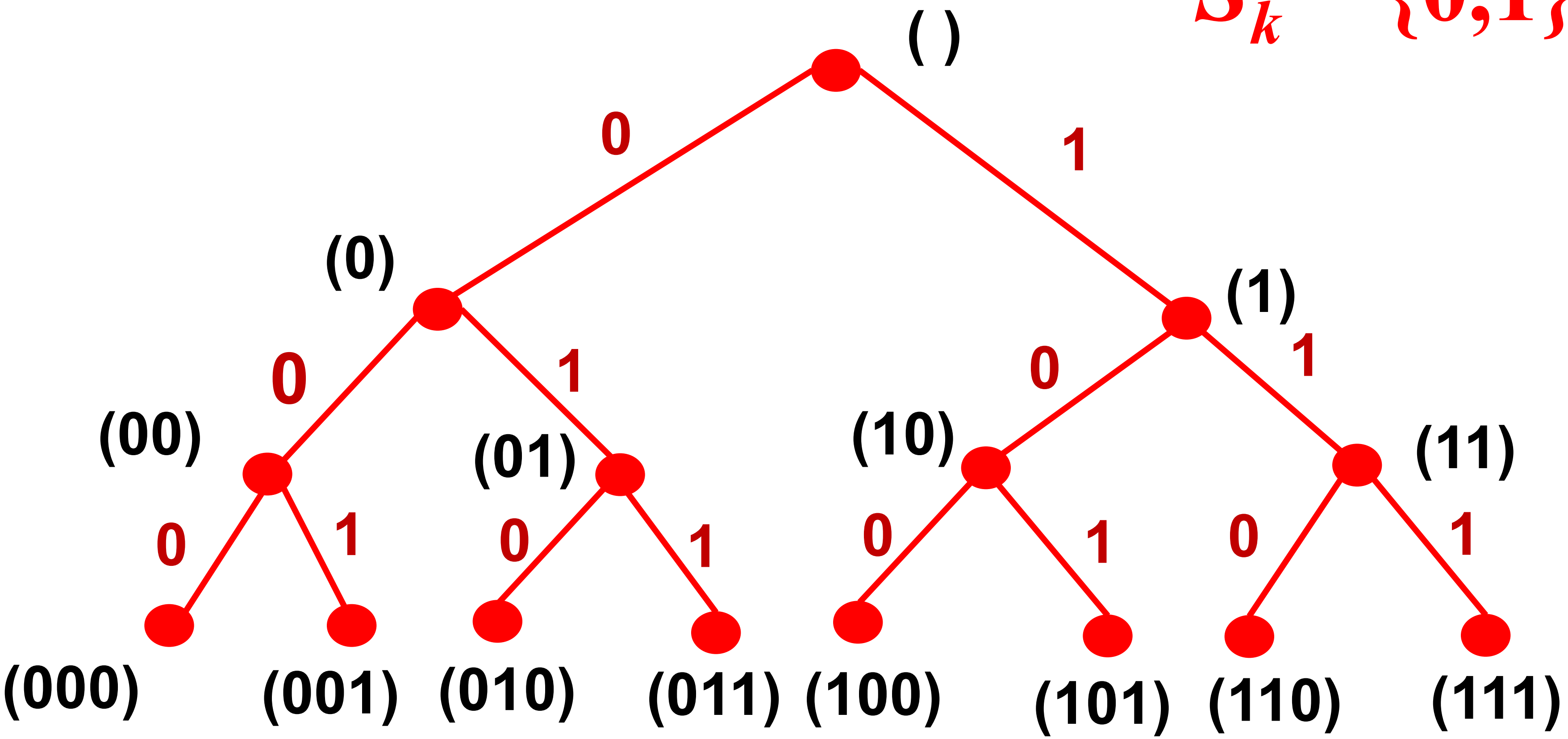
- We consider how to solve two issue keys to implement backtracking algorithm:
  - Build candidate set S_k: We have $S_1 = \{0, 1\}$. Assume we have binary string of length k-1 $(a_1, ..., a_{k-1})$, then $S_k = \{0,1\}$. Thus, the candidate sets for each position of the solution are determined.
  - Implement the loop to enumerate all elements of S_k: we can use the loop for

    for (y=0; y<=1; y++) in C/C++

Generate binary strings of length *n*

Decision tree:

$$S_k = \{0,1\}$$



Generate binary strings of length *n*

```
#include <iostream>
using namespace std;
int  n, count;
int a[100];

void PrintSolution()
{
  int i, j;
  count++;
  cout<<"String # " <<count<<": ";
  for  (i=1 ; i<= n ;i++)
  {
      j=a[i];
      cout<<j<<"   ";
  }
  cout<<endl;
}
```

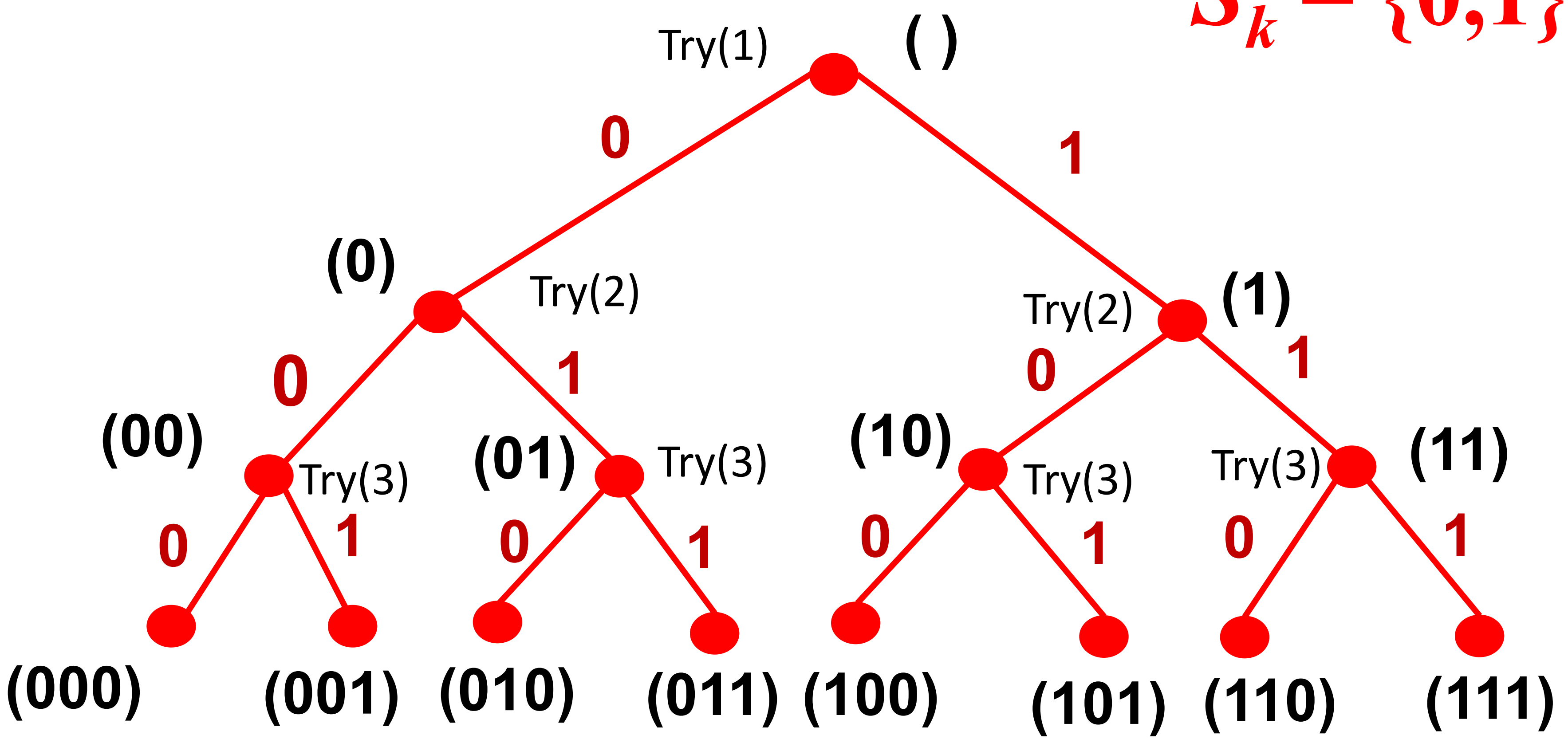
```
void  Try(int k)
{
   for (int j = 0; j<=1; j++)
   {
       a[k] = j;
       if (k == n)   PrintSolution();
       else  Try(k+1);
    }
}

int main()
{
   cout<<"Enter n = ";cin>>n;
   count = 0; Try(1);
   cout<<"Number of strings "<<count;
}
```

Generate binary strings of length *n*

Decision tree:

$$S_k = \{0,1\}$$



Generate binary strings of length *n*

```
#include <iostream>
using namespace std;
int  n, count,k;
int a[100], s[100];

void PrintSolution()
{
     int i, j;
     count++;
     cout<<"String # " << count<<": ";
     for  (i=1 ; i<= n ;i++)
         cout<<a[i]<<"   ";

      cout<<endl;
}
```

```
void  GenerateString( )
{
      k=1;  s[k]=0;
      while (k > 0)
     {
           while (s[k] <= 1)
           {
                 a[k]=s[k];
                 s[k]=s[k]+1;
                 if (k==n)  PrintSolution();
                 else
                 {
                      k++; s[k]=0;
                  }
            }
            k--;   // BackTrack
      }
}
```

Generate binary strings of length *n*

```
int main() {
      cout<<"Enter value of  n = ";cin>>n;
      count = 0; GenerateString();
      cout<<"Number of strings = "<<count<<endl;

}
```

3.1. Algorithm diagram

3.2. Generate basic combinatorial configurations

- Generate binary strings of length n
- **Generate m-element subsets of the set of n elements**
- Generate permutations of n elements

Generate *m*-element subsets of the set of *n* elements

Problem: Enumerate all m-element subsets of the set n elements $N = \{1, 2, ..., n\}$.

Example: Enumerate all 3-element subsets of the set 5 elements N = {1, 2, 3, 4, 5}

Solution: (1, 2, 3), (1, 2, 4), (1, 2, 5), (1, 3, 4), (1, 3, 5), (1, 4, 5), (2, 3, 4), (2, 3, 5), (2, 4, 5), (3, 4, 5)

➔ Equivalent problem: Enumerate all elements of set:

$$S(m,n)=\{(a_1,\ldots, a_m)\in N^m: 1 \le a_1<\ldots<a_m\le n\}$$

Generate m-element subsets of the set of n elements

We consider how to solve two issue keys to implement backtracking:

- Build candidate set S_k:
  - With the condition: : $1 \le a_1 < a_2 < ... < a_m \le n$

  we have $S_1 = \{1, 2, ..., n-(m-1) \}$.
  - Assume the current subset is $(a_1, ..., a_{k-1})$, with the condition $a_{k-1} < a_k < . . . < a_m \le n$, we have $S_k = \{a_{k-1}+1, a_{k-1}+2, ..., n-(m-k)\}$.
- Implement the loop to enumerate all elements of S_k: we can use the loop for

  for (j=a[k-1]+1;j<=n-m+k;j++)..

Generate *m*-element subsets of the set of *n* elements

```
#include <iostream>
using namespace std;

int  n, m, count;
int a[100];
void PrintSolution() {
    int i;
    count++;
    cout<<"The subset #" <<count<<": ";
    for  (i=1 ; i<= m ;i++)
          cout<<a[i]<<"  ";
    cout<<endl;
}
```

```
void  Try(int k){
   int j;
   for (j = a[k-1] +1; j<= n-m+k; j++) {
         a[k] = j;
         if (k==m)  PrintSolution();
         else Try(k+1);
    }
}
int main() {
   cout<<"Enter n, m = "; cin>>n; cin>>m;
   a[0]=0; count = 0; Try(1);
   cout << "Number of " << m;
   cout << "-element subsets of set " << n;
   cout << " elements = " << count << endl;
}
```

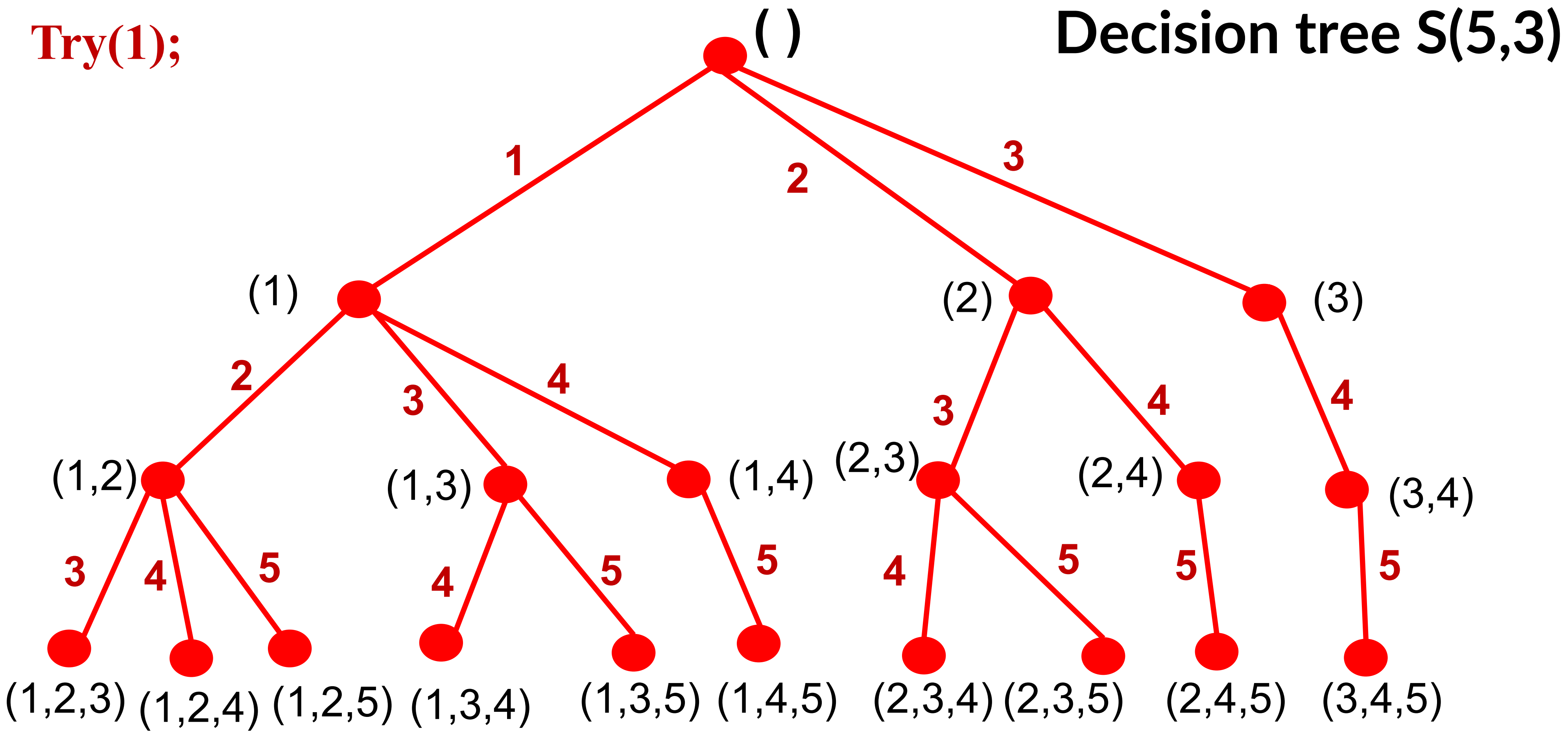
Generate *m*-element subsets of the set of *n* elements

```
#include <iostream>
using namespace std;

int  n, m, count,k;
int a[100], s[100];
void PrintSolution() {
    int i;
    count++;
    cout<<"The subset #" <<count<<": ";
    for  (i=1 ; i<= m ;i++)
         cout<<a[i]<<"  ";
    cout<<endl;
}
```

```
void  MSet()
{
    k=1;  s[k]=1;
    while(k>0){
        while (s[k]<= n-m+k) {
            a[k]=s[k];  s[k]=s[k]+1;
            if (k==m) PrintSolution();
            else { k++; s[k]=a[k-1]+1;  }
        }
        k--;
    }
}
int main() {
   cout<<"Enter n, m = "; cin>>n; cin>>m;
   a[0]=0; count = 0; MSet();
   cout<<"Number  of  "<<m<<"-element  subsets  of  set  "<<n<<"
   elements = "<<count<<endl;
}
```

Generate *m*-element subsets of the set of *n* elements



$$S_k = \{a_{k-1}+1,\ a_{k-1}+2,\ \ldots,\ n-(m-k)\}$$

3.1. Algorithm diagram

3.2. Generate basic combinatorial configurations

- Generate binary strings of length *n*
- Generate *m*-element subsets of the set of *n* elements
- **Generate permutations of *n* elements**

Enumerate permutations

Permutation set of natural numbers 1, 2, ..., n is the set:

$$\Pi_n = \{(x_1,\dots, x_n) \in N^n:\ x_i \neq x_j\ ,\ i \neq j\ \}.$$

Problem: Enumerate all elements of Π_n

Enumerate permutations

- Build candidate set S_k:
  - Actually $S_1 = N$. Assume we have current partial permutation $(a_1, a_2, ..., a_{k-1})$, with the condition $a_i \neq a_j$, for all $i \neq j$, we have

    $$S_k = N \setminus \{ a_1, a_2, ..., a_{k-1}\}.$$

Enumerate permutations

- Describe set S_k: **Build function to detect candidates:**

```
bool candidate(int j, int k) {
  //function returns true if and only if j ∈ Sk
   int i;
   for (i=1;i++;i<=k-1)
       if (j == a[i]) return false;
   return true;
}
```

Enumerate permutations

- Implement the loop to enumerate all elements of S_k:

```
for (j=1; j <= n; j++;)
   if  (candidate(j, k)) {
         // j is candidate for position kth
         . . .
   }
```

Enumerate permutations

```
#include <iostream>
using namespace std;

int  n, m, count;
int a[100];

int PrintSolution() {
    int i, j;
    count++;
    cout<<“Permutation #"<<count<<": ";
    for  (i=1 ; i<= n ;i++)
          cout<<a[i]<<"  ";
    cout<<endl;
}
```

```
bool candidate(int j, int k){
    int i;
    for (i=1; i<=k-1; i++)
       if (j == a[i])
              return false;
    return true;
}
```

Enumerate permutations

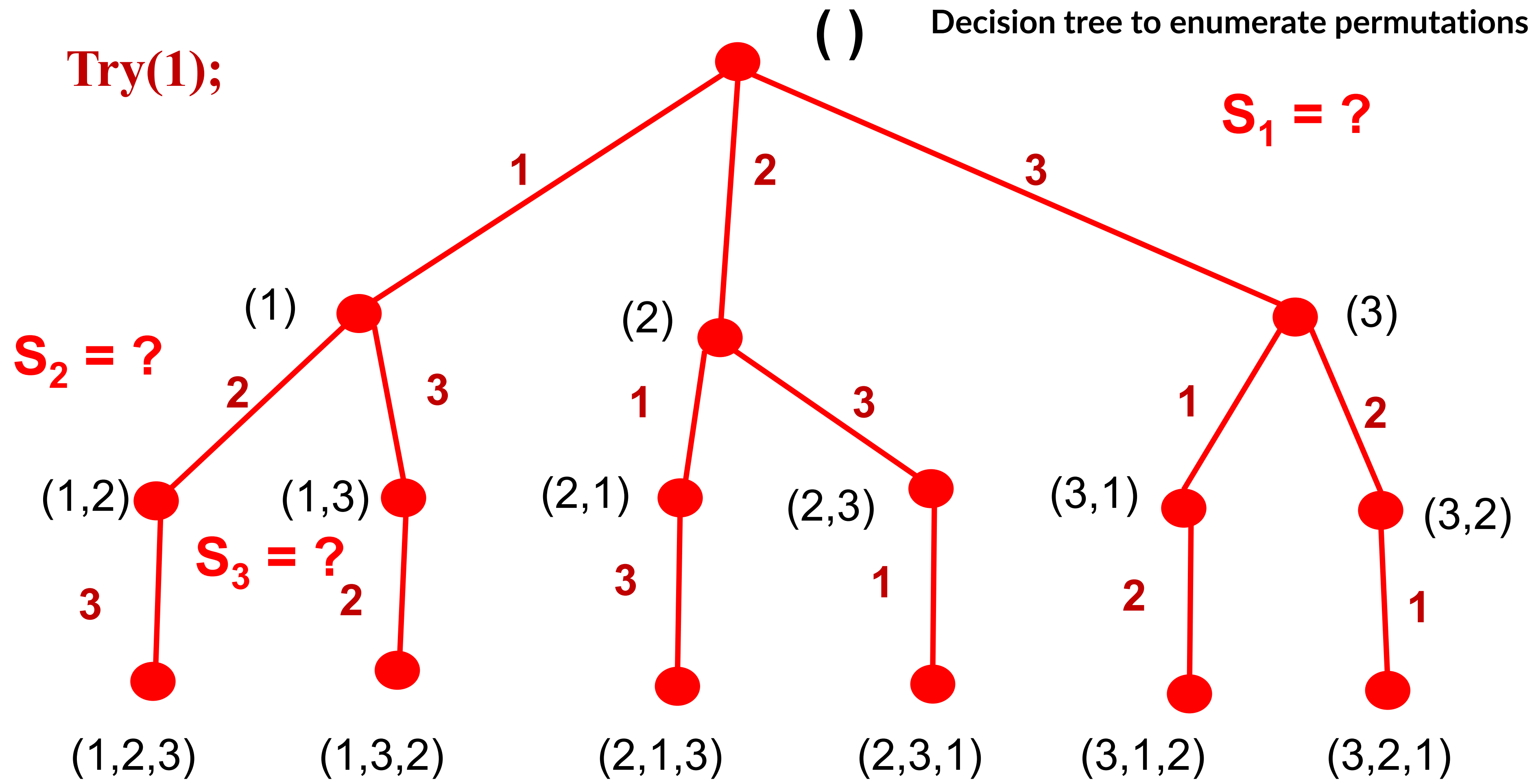
```
void  Try(int k) {
   int j;
   for (j = 1; j<=n; j++)
      if (candidate(j,k)) {
         a[k] = j;
         if (k==n)  PrintSolution( );
         else  Try(k+1);
      }
}
```

```
int main() {
    cout<<("Enter n = "); cin>>n;
    count = 0; Try(1);
    cout<<"Number of permutations = " << count;
}
```

Enumerate permutations

Try(1);

Decision tree to enumerate permutations of 1, 2, 3



$$S_k = N \setminus \{ a_1, a_2, \ldots, a_{k-1} \}$$

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THANK YOU !